Research on Innovation Path of 3D Transistor Architecture

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Abstract. The exponential growth of the integrated circuit (IC) industry, historically governed by Moore's Law, now faces challenges as traditional silicon based planar MOSFETs approach their fundamental physical limits. Among all solutions to sustain IC development, three-dimensional structural innovations in transistor architecture present a key viable pathway. This paper analyzes the evolution from planar structures to the three leading variants that have shown successful progress and promising potential: Fin Field-Effect Transistors (FinFET), Gate-All-Around FETs (GAAFET), and Complementary Field-Effect Transistors (CFET). Each variant has unique strengths and exists at a different stage of maturity and industry adoption. Through its tri-gate design, FinFET has mitigated leakage and improved electrostatic control. Through its gate-all-around design, GAAFET offers enhanced electrostatic control, which boosts on-current by reducing quantum capacitance effects and improving carrier mobility. Looking ahead to beyond the 1nm node, CFET vertically stacks n-type and p-type transistors and promises unprecedented reductions in circuit footprint. However, while FinFET nears its limits and GAAFET emerges as its successor, future efforts must tackle CFET's key manufacturing and thermal hurdles to enable further advances.

Keywords: Integrated Circuit; MOSFETs; FinFET; GAAFET; CFET.

1. Introduction

Integrated circuits (ICs) drive IT functionality and computing efficiency [1] while following Moore's Law. This principle predicts that the number of integrated complementary metal-oxide-semiconductor (CMOS) transistors will double every 12 to 18 months [2]. The reduction of transistor size, which allows for higher transistor density, is the key to their versatility, made possible by advancements in controlling carrier transport—particularly in silicon (Si)-based semiconductor devices [3, 4].

However, as scientists and engineers continue to advance the currently dominant and mature semiconductor technology, which relies on CMOS manufacturing for ICs, new challenges have emerged. They recognize that transistors can only be scaled down so far, and eventually, traditional silicon MOSFET-based transistors will no longer meet the demands of miniaturization. There is an urgent need to overcome the limitations caused by scaling, such as threshold voltage scaling, gate oxide tunneling, parasitic resistance and capacitance, and short channel effects [5]. Fig. 1 shows the leakage current of MOSFETs.

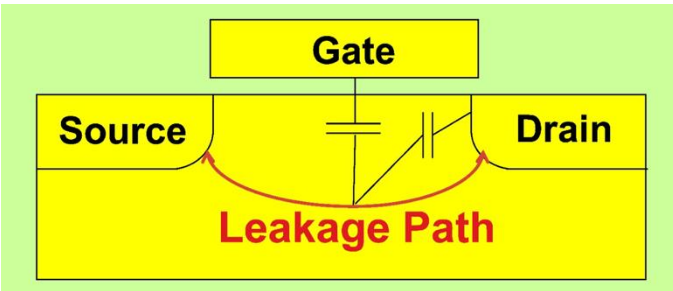


Fig. 1 Schematic diagram of leakage current [6]

Particularly in 2025, AI and database-driven applications—where GPUs and specialized accelerators demand unprecedented transistor performance—are driving stricter requirements for integrated circuits (ICs). Therefore, the urgency of breaking through traditional silicon-based MOSFETs was emphasized. This paper explores both current and emerging solutions for how structural innovations can address these challenges. By synthesizing recent advancements, this work aims to highlight pathways for sustaining the computing efficiency and functionality of ICs while simultaneously enabling research on next-generation ICs capable of supporting the evolving exigencies of future technologies.

2. Structural Innovations

Structural innovation is a critical pathway that engineers have continuously pursued. Without structural innovation, Moore’s Law would have ended over a decade ago [7]. The structure of CMOS technology consists of the Source, Gate, and Drain on a silicon wafer as shown in Fig. 2. It operates by controlling the gate voltage to adjust the mobility of charge carriers in the channel. [1].

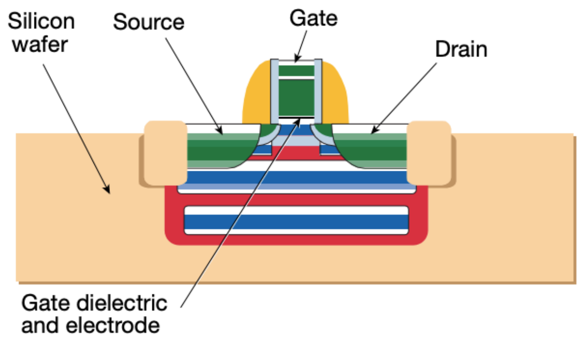


Fig. 2 Structural diagram of planar metal oxide semiconductor (MOS) transistor [8]

In order to downscale planar CMOS transistors by modifying their structure, the thickness of the gate dielectric—traditionally made of silicon dioxide—must decrease as the gate length decreases [8]. This makes the thickness of silicon dioxide a key challenge that needs to be addressed [8].

If silicon dioxide is continued to be used as the gate insulator in the planar structure of the 35nm technology node, the thickness will be reduced to 0.5nm (approximately only two atomic layers) [8]. Reducing the oxide thickness below ~1.5 nm results in unacceptable performance due to gate tunneling leakage and dielectric breakdown [8], making this approach unsuitable for further downscaling.

As a result, engineers have moved away from traditional planar structures as shown in Fig. 3. Fin field-effect transistors (FinFET) and gate surround field-effect transistors (GAAFET) have been proposed, which can better control short channel effects (SCE), low junction leakage, and high carrier mobility [9]. Looking further ahead, CFETs (Complementary FETs)—with a critical dimension of only 14 Å (1 Å = 0.1 nm)—are expected to be commercially introduced after 2031 [2].

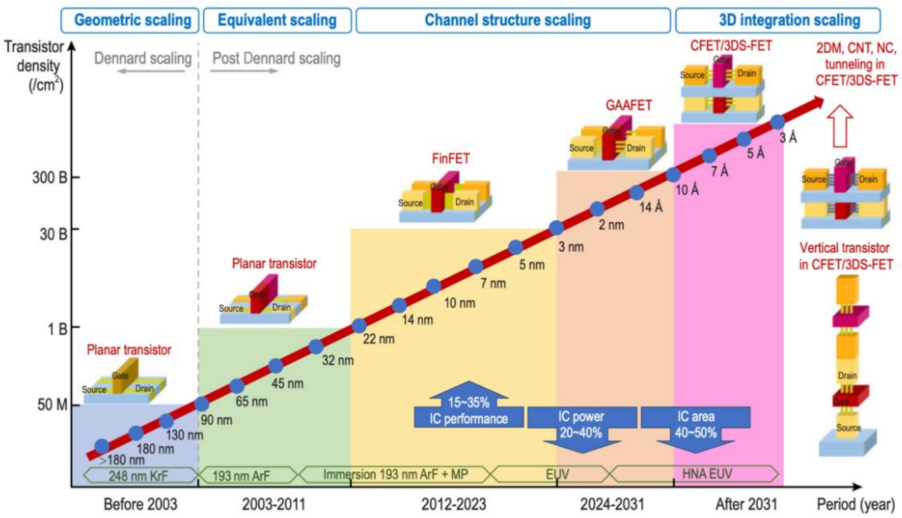


Fig. 3 Historical development and future trends in MOSFET transistor structures and scaling [2]

2.1. FinFET

The most successful solution to date is the FinFET—a 3D fin-shaped transistor architecture proposed by Hu Chenming's team in the 1990s. By the early 2010s, FinFETs had begun replacing the traditional planar transistors that had dominated the semiconductor industry for decades. A FinFET is structurally distinct from a planar MOSFET due to its vertical, fin-shaped channel erected on the insulating substrate with the source and drain at either end [10]. Unlike a planar transistor, the FinFET's gate surrounds the channel on three sides, as shown in Fig. 4.

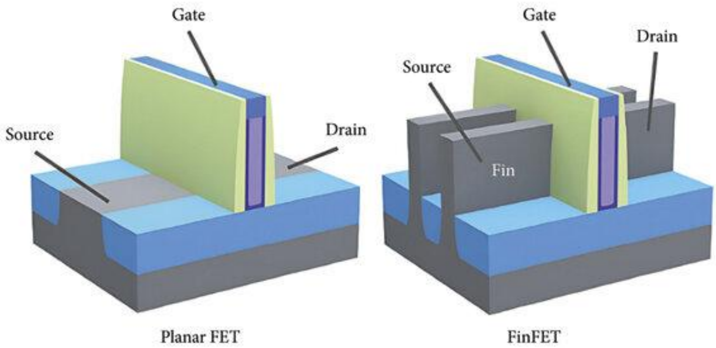


Fig. 4 Comparison of FinFET and planar FET [10]

The Fin like structure of FinFET directly reduces leakage current [10] and mitigates short channel effects. This approach allows for better control of current, reduces leakage, and improves switching speed [11]. Furthermore, the FinFET's channel requires minimal doping, which reduces interference from impurity ions and enhances carrier mobility – the combined movement rate of electrons and holes within the semiconductor [12].

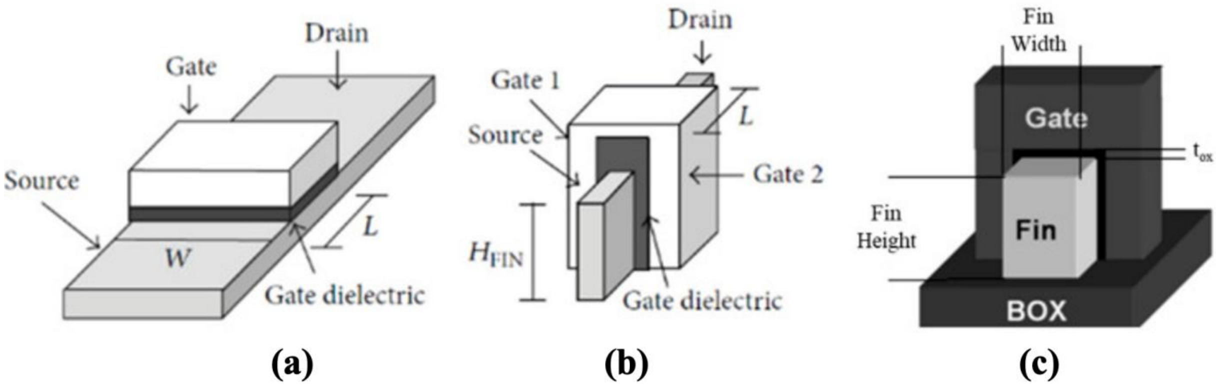


Fig. 5. Structure schematic of (a) planar MOSFET and (b) FinFET, (c) Demonstrating H_{Fin} as fin height and W_{Fin} as fin width [13]

However, under further device miniaturization beyond the 5-nm node, leakage and short-channel effects will reappear in FinFETs [10]. The biggest shortcoming of FinFETs is parasitic capacitance. Due to the increased overlap area between the front and back gates [13], the capacitances of the devices increase [14] (The structure is shown in Figure 5). To alleviate these parasitic capacitances, increasing the fin height (H_{Fin}) while reducing the fin pitch (FP) is an effective method. as shown in Fig. 6 [13, 14, 15]. However, at the required extreme aspect ratios, the tall, narrow fin structure becomes susceptible to deformation [16]. It can bend or warp due to mechanical stress from the surrounding materials and localized heating [16]. This deformation risks causing electrical shorts and renders consistent manufacturing with high yield impossible. This fundamental limit to FinFET scaling has pushed engineers to focus on next-generation gate-all-around (GAA) architectures [1].

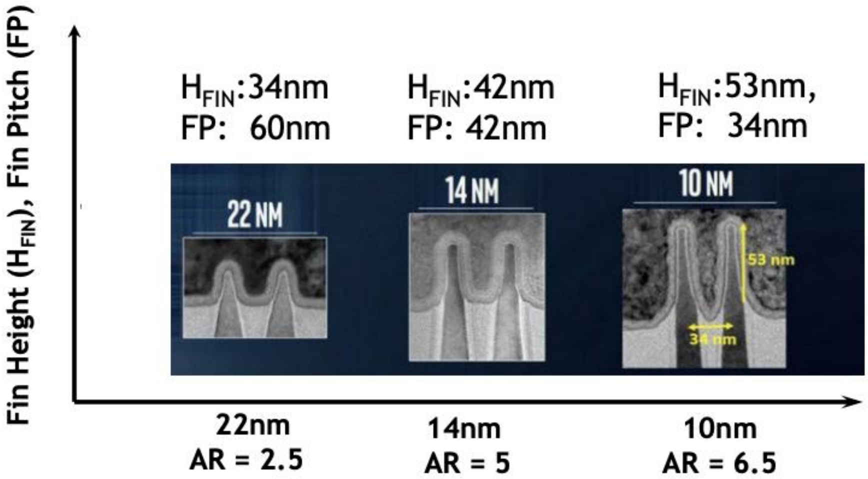
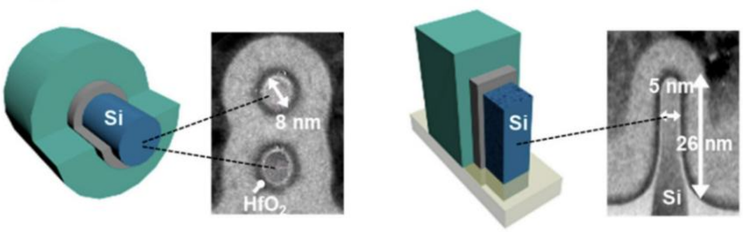


Fig. 6. The increase of H_{Fin} and decrease of FP from 22nm to 10nm node [14]

2.2. GAAFET

A key improvement emerged in the form of GAA architectures, where engineers typically replace the vertical fin with stacked nanosheets (NS) or nanowires (NW) [17]. These individual horizontal nanosheets are vertically stacked, allowing the gate to surround the channel on all four sides—in contrast to the earlier FinFET design, where the gate only wraps around three sides [17]. This full gate enclosure offers better electrostatic control, further reducing leakage and increasing drive current [18, 19]. A critical design flexibility of GAA architectures is the ability to tune the width of the nanosheets: wider sheets maximize the drive current for performance-critical circuits, while narrower sheets can optimize for lower power consumption, as shown in Fig. 7 [17]. This tunability enables more precise design-level optimization of power and performance.



(a) **(b)**

Fig. 7. Cross sectional images of (a) GAA NW-FETs and (b) FinFETs [17]

The enhanced electrostatic control provided by the gate around the channel enables GAAFET to have high driving current. GAAFET exhibits a higher on current due to the effective control of the entire channel by the gate. This reduces the effects of quantum capacitance and improves carrier mobility [17].

Thanks to their immense advantages, GAAFETs are gaining widespread adoption. Compared to its 5nm node, Samsung's 3-nanometer process node reduces power consumption by 45% and surface area by 16% with its GAA technology. In Samsung’s latest mobile processor, the Exynos 2500 (used in its latest Galaxy Series smartphones), GAAFET technology has enabled better power efficiency and enhanced heat dissipation while its fan-out wafer-level packaging (FOWLP) allows for a much-reduced chip thickness [20, 21]. AMD's latest "Venice" EPYC server processor is one of the first high-performance computing products to adopt TSMC N2 technology. Compared to the previous N3 process, at the same speed, power consumption will be reduced by 25% -30%, and chip density will be increased by 15% [20].

2.3. CFET

Complementary FET (CFET), or the 3D stacked FET (3DS FET), has been shown to be even more promising as a next-generation architecture beyond the 1-nm node. CFET architecture overcomes the limitations of horizontal n-to-p spacing by vertically stacking the n-type and p-type transistors [1]. The folding of n-channels and p-channels on a single package enables extremely compact standard cell designs with 4T or even 3T track heights [1, 22].

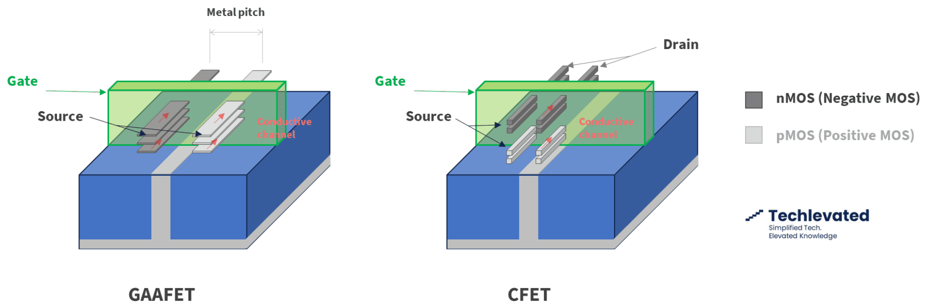


Fig. 8. Vertical stacking structure of a CFET [20].

By utilizing the vertical dimension, this stacked configuration allows for a greater total effective channel width within the same footprint, which thereby enhances the drive current. Furthermore, fabricating such devices using a junction-less (JL) poly-Si process at low temperatures (≤600 °C) is highly beneficial for monolithic 3D ICs and system-on-panel (SoP) applications, as it offers a simple process with a low thermal budget [23]. Experimental results demonstrate that a vertically stacked structure can more than double the on-current compared to a single nanosheet FET [23].

However, a key challenge for the CFET inverter is the independent adjustment of the nFET and pFET electrical characteristics. Since the pFET is stacked over the nFET and the process flow requires both devices to have an identical channel width [24], varying this channel width parameter simultaneously shifts the threshold voltages and currents of both transistors—making it difficult to achieve symmetric on-current matching and optimal voltage transfer characteristics (VTCs) for circuit operation [23, 24].

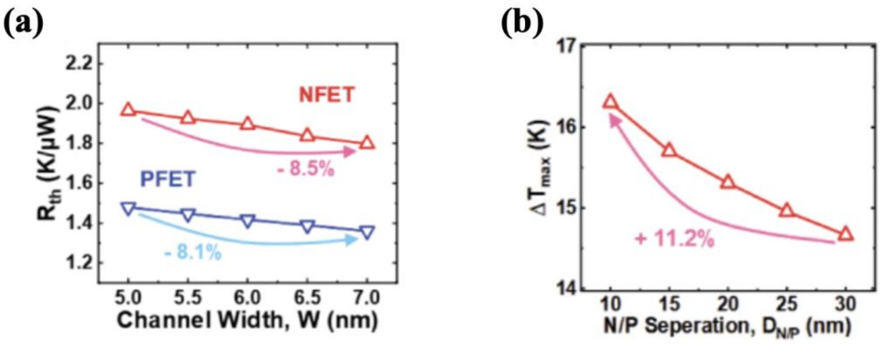


Fig. 9. (a) R_{th} of top n-nanosheet FET and bottom p-FinFET devices. (b) ΔTmax [24]

Furthermore, this monolithic integration creates severe electrothermal coupling challenges [24]. The top device exhibits a higher thermal resistance (Rth = 1.89 K/μW, as seen in Fig. 9) than the bottom device (Rth = 1.42 K/μW). Although reducing the vertical spacing of nFET/pFET from 30 nm to 10 nm resulted in an 11.9% faster operating frequency and a 3.4% reduction in dynamic power loss, it concurrently increased the lattice temperature (by 11.2%) for the entire stack [24]. This fundamental electrothermal coupling requires a careful level of design-technology co-optimization to balance electrical performance against thermal reliability [24]. Consequently, CFETs remain an active research area and have not yet matured to a stage suitable for high-volume commercial production.

3. Conclusions

The relentless pursuit of Moore's Law has driven the IC industry to the physical limits of conventional silicon-based planar MOSFETs. To address these challenges, structural innovations have thus become essential. Specifically, three-dimensional transistor architectures have been pivotal. Within this context, three leading variants have shown successful progress and promising potential: FinFET, GAAFET, and CFET.

FinFET is a successful approach that has extended the life of CMOS technology for over a decade. With its tri-gate design, FinFET successfully mitigates leakage and short-channel effects at nodes from 22 nm to 5 nm. However, its scalability is constrained by parasitic capacitance and the mechanical instability of ultra-scaled fins. This has then catalyzed the industry's shift toward GAAFETs. GAAFETs utilize stacked nanosheets or nanowires, with the gate surrounding the channel on all four sides. They provide even superior electrostatic control, higher drive currents, and greater power efficiency, while simultaneously offering greater design flexibility through tunable channel widths. This technology has been successfully researched and developed, as evidenced by the cutting-edge 3 nm processes from leading foundries like Samsung and TSMC.

Looking beyond the 1 nm node, CFETs vertically stack nFET and pFET devices—and this vertical stacking promises an unprecedented reduction in standard cell size. They can potentially enable 4-track or even 3-track libraries, offering a revolutionary solution to the interconnect bottleneck and density scaling challenges. However, this architecture is still new and immature, as it faces complexities in achieving independent threshold voltage tuning and managing severe electrothermal coupling between the vertically stacked transistors (the top device has a higher thermal resistance (Rth = 1.89 K/μW) than the bottom device (Rth = 1.42 K/μW), and the maximum lattice temperature increase (ΔTmax) is 11.2% higher). The trade-off between performance gains and thermal reliability underscores the need for design-technology co-optimization.

While the era of simple planar scaling has ended, the path forward for Moore's Law is being forged through architectural ingenuity. The progression from FinFET to GAAFET to CFET demonstrates that the physical limits of one generation can be overcome by reimagining the transistor's form factor in three dimensions. However, in practice, FinFET will soon approach its scaling limits, just as planar MOSFETs did, and GAAFET is emerging as FinFET's imminent successor. Thus, future research must focus on overcoming the manufacturing and thermal challenges of CFETs. Exploring novel materials to complement these new structures and developing sophisticated co-design frameworks to balance performance, power, and reliability could be key solutions. The continued evolution of transistor architecture will remain indispensable for powering the next generation of computing.

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